

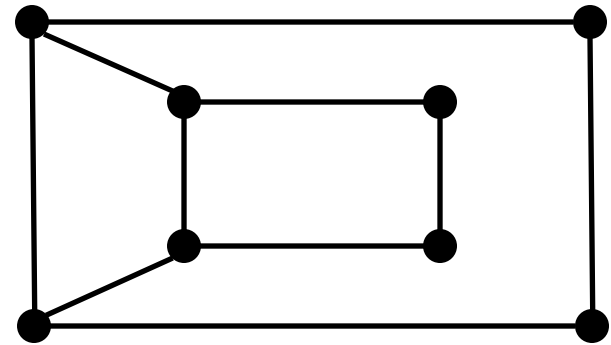
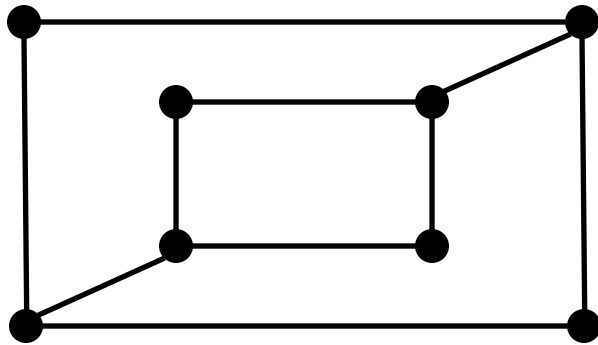
Bipartite Graphs

Previous Lecture

- *Graph representations*
 - *Vertex/edge lists, adjacency lists, adjacency matrices, incidence matrices*
- *Graph isomorphism*
- *Graph invariants*
- *Subgraphs*

Error from last class

Example: Are these graphs isomorphic?



No!

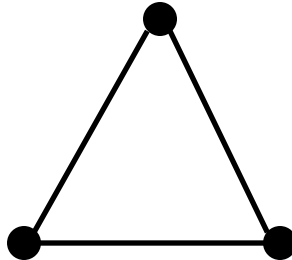
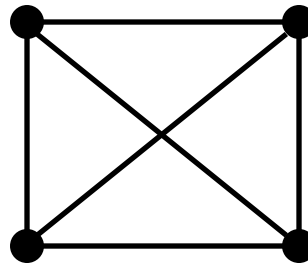
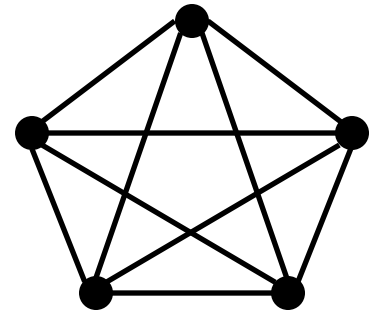
Right one has 3 cycles of length 4, but the left only has 2 of length 4

Left has 4 cycles of length 6, but the right only has 2

Right has a cycle of length 8, left does not

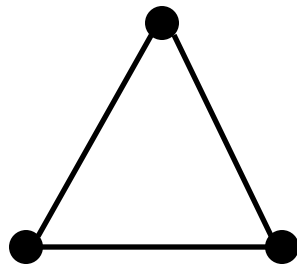
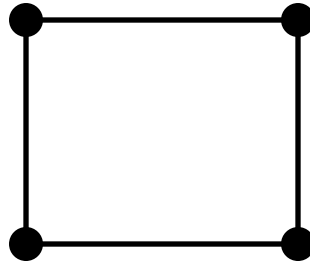
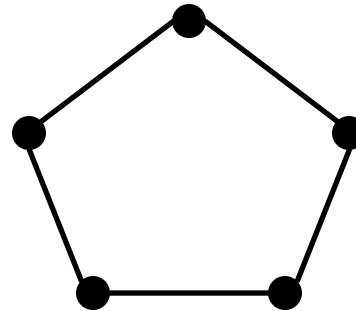
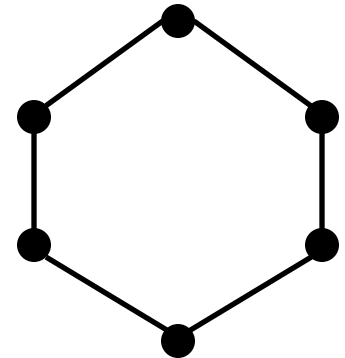
Some Special Graphs

- **Complete Graphs.** *The complete graph K_n on n vertices is the simple graph that contains exactly one edge between each pair of distinct vertices*

 K_1  K_2  K_3  K_4  K_5

Some Special Graphs (cntd)

● **Cycles.** The cycle C_n for $n \geq 3$ consists of n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$.

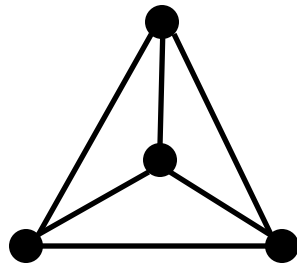
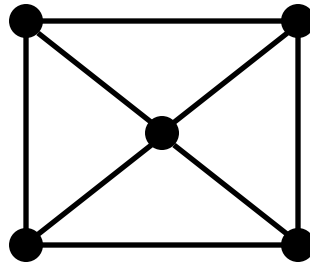
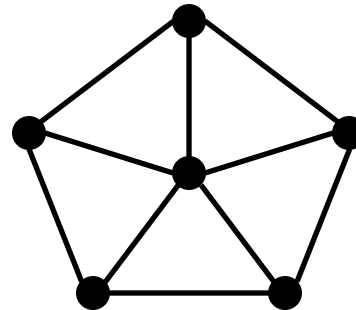
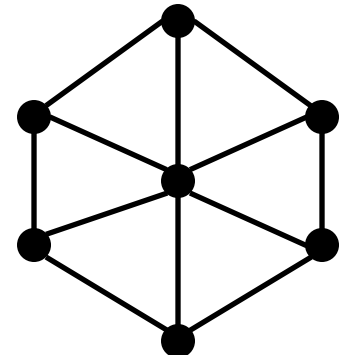
 C_3  C_4  C_5  C_6

● **Theorem:**

A graph has a loop of length k if and only if it has a subgraph isomorphic to C_k

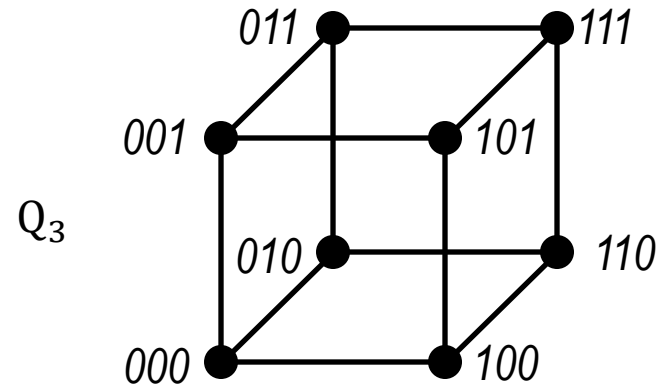
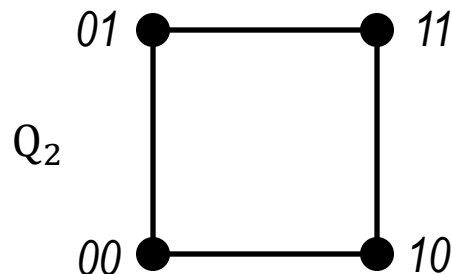
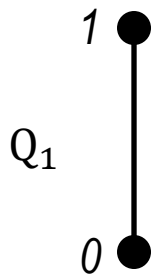
Some Special Graphs (cntd)

- **Wheels.** The wheel, w_n , is obtained from the cycle c_n when we add an additional vertex to it and connect this vertex to each of the n vertices in c_n by new edges

 W_3  W_4  W_5  W_6

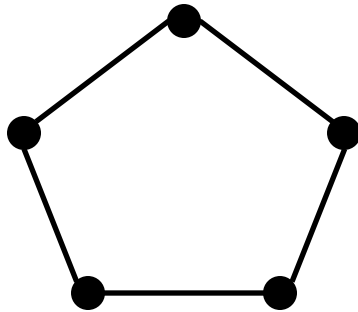
Some Special Graphs (cntd)

- **Hypercubes.** The n -dimensional hypercube, or n -cube, denoted by Q_n is the graph that has vertices representing the bit strings of length n . Two vertices are adjacent if and only if the bit strings that they represent differ in exactly one bit position.

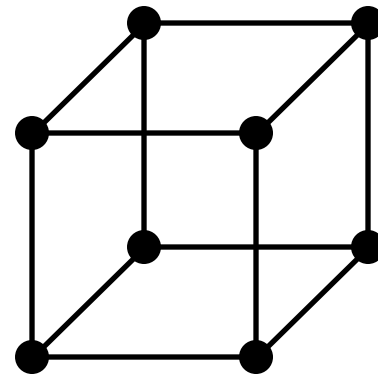


Regular Graphs

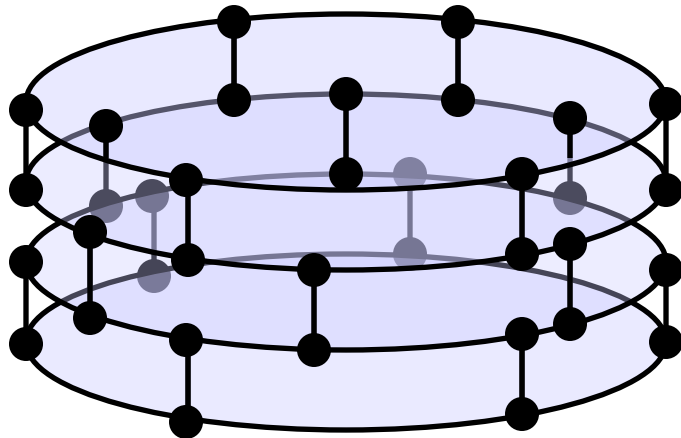
● A graph is called *k-regular* if all its vertices have degree k .



2-regular graphs are cycles



The k-cube is a k-regular graph



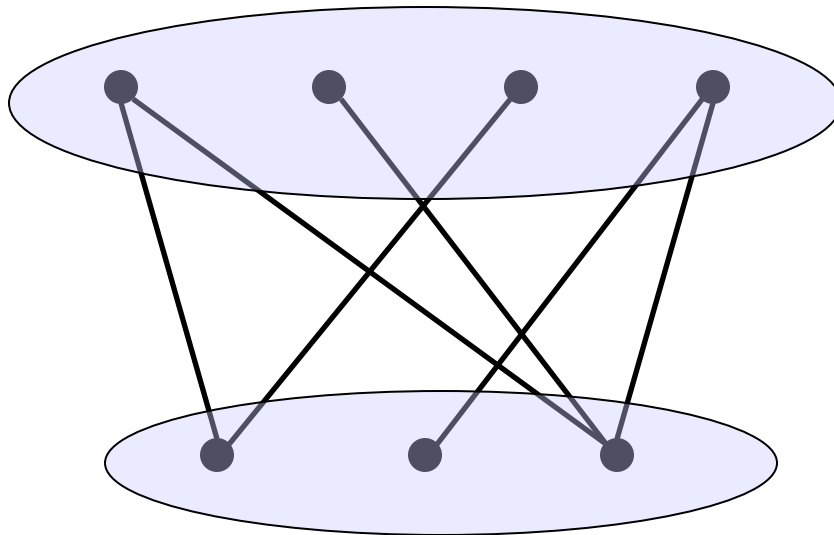
A brick graph, another example of a 3-regular graph

Are they isomorphic?

Bipartite Graphs

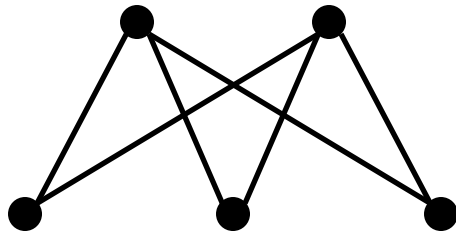
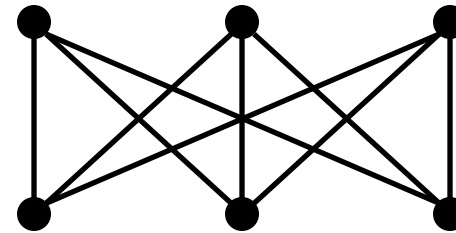
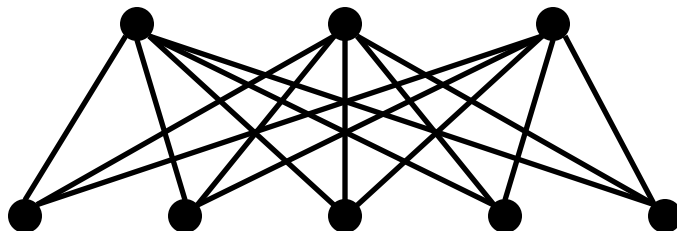
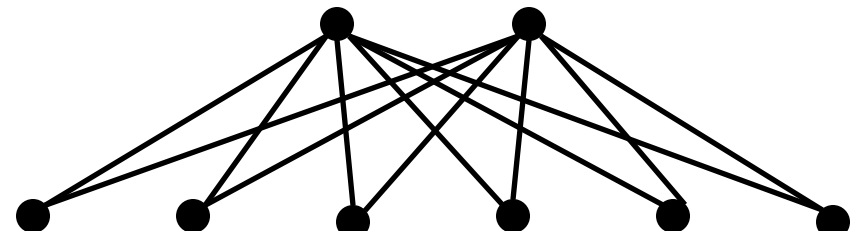
- A simple graph G is **bipartite** if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 .

When this condition holds, we call the pair V_1, V_2 a **bipartition** of the vertex set V of G



Complete Bipartite Graphs

- The complete bipartite graph $K_{m,n}$ is the graph that has its vertex set partitioned into two subsets of m and n vertices, respectively. There is an edge between two vertices if and only if one vertex is in the first subset and the other vertex is in the second subset.

 $K_{2,3}$  $K_{3,3}$  $K_{3,5}$  $K_{2,6}$

More Bipartite Graphs

● Theorem.

For a simple graph $G = (V, E)$ the following conditions are equivalent:

- (1) G is bipartite,*
- (2) G does not contain a cycle of odd length,*
- (3) it is possible to color each vertex of G in one of two different colors so that no two adjacent vertices are colored the same color.*

● Proof:

We prove implications $(1) \rightarrow (2)$, $(2) \rightarrow (3)$, and $(3) \rightarrow (1)$.

More Bipartite Graphs (cntd)

● (1) $\rightarrow$ (2) (*bipartite implies no odd cycles*)

Suppose that G is bipartite, and U, W is a bipartition.

Let $u = u_0, e_1, u_1, \dots, e_{n-1}, u_{n-1}, e_n, u_n = u$ be a cycle. Without loss of generality we may assume that $u \in U$.

Then $u_1 \in W, u_2 \in U, u_3 \in W, \dots$ and in general, u_i is in U if and only if i is even.

But $u_n = u$ is in U , so n is even as required.

More Bipartite Graphs (cntd)

● (2) $\rightarrow$ (3) (no odd cycles implies 2-colourable)

Suppose that every cycle in G has even length.

We show that the following algorithm constructs a proper coloring of G in two colors

take any vertex and **color** it blue

repeat the following **while** there is an uncolored vertex

take any vertex v that is already colored

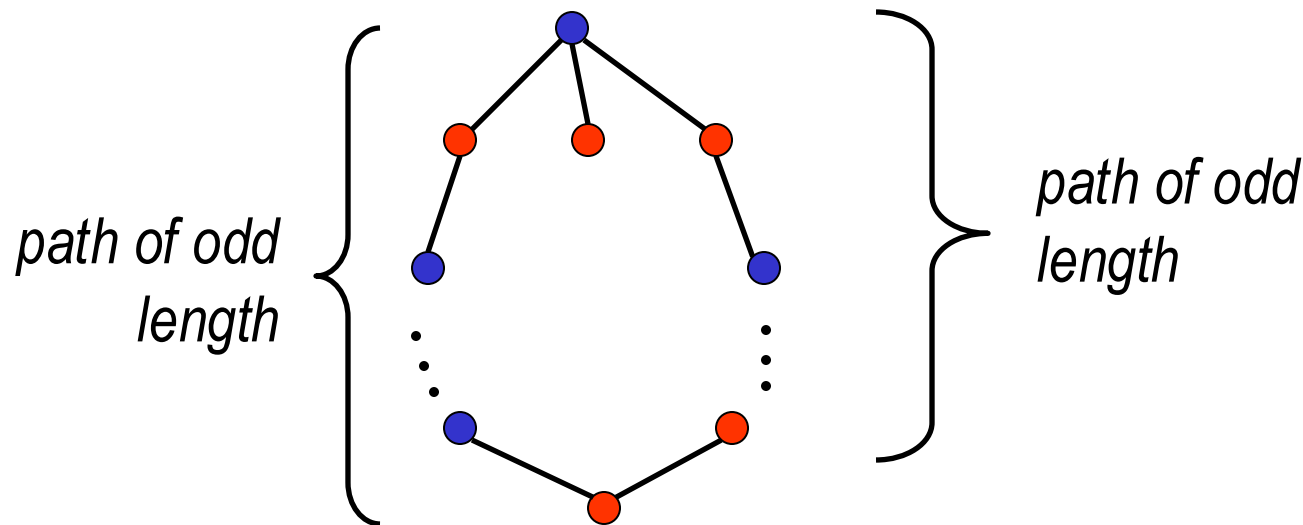
if v is colored blue **then color** all its neighbors red

if v is colored red **then color** all its neighbors blue

end repeat

More Bipartite Graphs (cntd)

- *To prove the correctness of this algorithm, we have to show that it never attempts to color red a vertex already colored blue, and that it never attempts to color blue a vertex already colored red*
- *Suppose that the algorithm attempts to color blue a vertex previously colored red*



- *Thus, we obtain a cycle of odd length; a contradiction.*

More Bipartite Graphs (cntd)

● (3) $\rightarrow$ (1) (2-colourable implies bipartite)

Suppose that the vertices of the graph G can be colored in two colors, red and blue, such that no two adjacent vertices receives the same color.

We construct a bipartition by including all blue vertices into one part of the bipartition, and all red vertices into the other part of the bipartition.

Q.E.D.

Practice

Exercises from the Book:

7th edition: 26, 53, 55 (page 666 – 667), 1, 3, 5, 7, 9c, 11, 39 (page 675 – 676)

8th edition: 26, 55, 57, (page 699 – 700), 1, 3, 5, 7, 9c, 11, 43 (page 710 – 712)